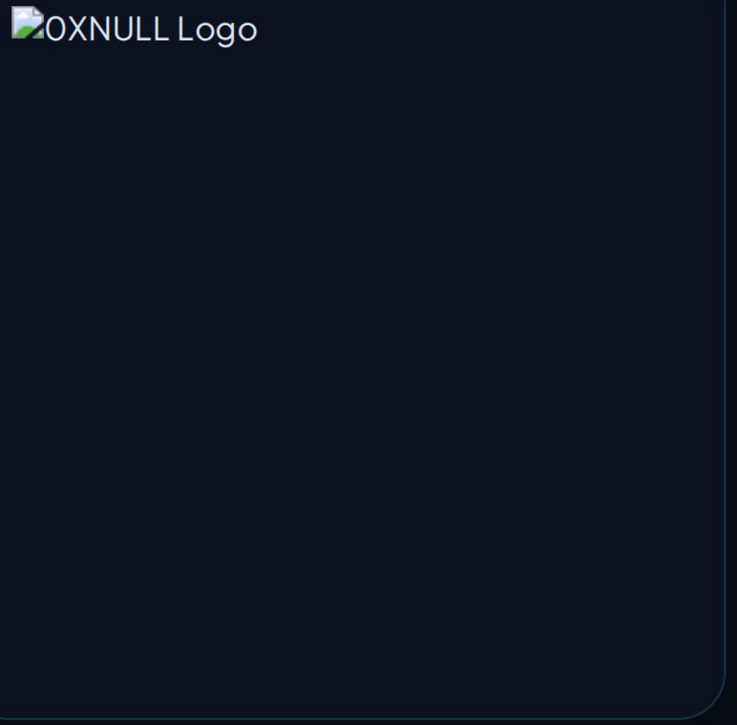
DEPARTMENT OF CYBERSECURITY

Club Proposal

0XNULL

A student-led academic initiative proposing a structured platform for cybersecurity learning, exploration, and practice — built under the Department of Cybersecurity.

CLUB — PROPOSAL



— PURPOSE

What the club helps students do.

01 / Build

Build strong common fundamentals

Anchor every student in shared cybersecurity foundations before specialising.

02 / Explore

Discover different security domains

Expose students to the breadth of cybersecurity — not just their major's track.

03 / Learn

Access curated learning resources

Maintain a vetted, evolving library of materials — papers, tools, walkthroughs.

04 / Share

Learn through peer-to-peer exchange

Students teach students — building communication and reinforcing understanding.

05 / Practice

Attend workshops & technical sessions

Recurring hands-on meetings that move from theory to demonstrable skill.

06 / Compete

Join activities and competitions

Members participate in CTFs, awareness drives, debates and showcases.

07 / Lead

Develop communication & leadership

Public speaking, presentation, event management and teamwork by running things.

08 / Update

Stay current on threats and the field

Threats, scams, technologies and industry developments through briefings.

09 / Discover

Discover possible career paths

Surface real industry roles, certifications, and growth directions with clarity.

— FOUNDATION

WHERE EVERY MEMBER BEGINS

Shared fundamentals, built before specialisation.

Whether a student arrives with strong systems knowledge or none, foundation provides a common starting floor so peer sessions stay focused and productive.

01

Computer fundamentals

Hardware, processors, memory, and binary computation.

02

Networking

TCP/IP, routing, DNS, and layered network traffic.

03

Operating systems

Processes, memory models, file systems, and kernel basics.

04

Linux

Command line, scripting, permissions, and toolchains.

05

Security fundamentals

Threat models, defence vocabulary, CIA triad, attack surfaces.

— SEVEN DOMAINS

After foundation — seven domains students explore.

Students engage with categories across Awareness, Offence–Defence, Forensics, Crypto, Cloud, and Architecture.

CATEGORIES
07

01 / Domain
Red & Blue Team
Offensive practice, defensive operations, and attack-defence exercises.

02 / Domain
Digital Forensics
Disk, memory, and network forensics investigating incidents.

03 / Domain
Cryptography
Symmetric, asymmetric, hashing, signatures and trust.

04 / Domain
Cloud Security
IAM, misconfigurations, container risks, and cloud posture.

05 / Domain
Security Architecture
Secure design, Zero Trust, threat modelling, defence-in-depth.

06 / Domain
Awareness
News, threats, scams, phishing, privacy, and emerging tech.

07 / Domain
Explore
Non-technical skills, leadership, PR, networking, and outreach.

Red & Blue Team

Two perspectives, one discipline — attack to understand, defend to protect. Every exercise is conducted in controlled labs with documented ethical boundaries.

RED TEAM

Offensive practice

Thinking from the attacker's position — finding weaknesses before threat actors do.

- Ethical hacking basics
- Reconnaissance techniques
- Vulnerability assessment
- Penetration testing
- Web application security
- Controlled attack simulation

BLUE TEAM

Defensive operations

Watching, detecting, responding — building the muscle that protects systems.

- Security monitoring
- Threat detection rules
- SIEM & log analysis
- Incident response drills
- Network perimeter defence
- Forensic triage

ACTIVITIES CTFs · structured labs · attack-defence exercises · debriefs

Digital Forensics

Reconstructing events from artefacts — disk, memory and network traces. The discipline of answering "what happened" with evidence, not assumption.

FOCUS AREAS

- Digital evidence handling
- Disk forensics
- Network forensics
- Incident investigation
- File systems analysis
- Memory forensics
- Log correlation & triage

EXAMPLE TOOLS COVERED

Wireshark

Packet capture and network traffic protocol analysis

Autopsy

Disk, image, and filesystem investigation platform

Volatility

Volatile memory artefact extraction and malware analysis

Cryptography

The mathematics that allows two parties to establish trust across an untrusted wire — turning plaintext into ciphertext impenetrable to adversaries.

FOCUS

- Encryption & decryption
- Asymmetric cryptography
- Digital signatures & PKI
- Symmetric ciphers (AES)
- Cryptographic hashing
- Key management & auth

WHERE THIS IS APPLIED

HTTPS & secure protocols	TL-1
Authentication & MFA systems	TL-2
Digital payment architectures	TL-3
Data at rest & transit protection	TL-4

Cloud Security

Most modern systems live in cloud infrastructure. Members learn the shared responsibility model, identity perimeters, and common misconfigurations.

01
Cloud fundamentals
Service models (IaaS/PaaS/SaaS) and shared responsibility.

02
Identity & access
IAM roles, least-privilege, and permission boundaries.

03
Cloud networking
VPCs, subnets, security groups, and ingress filtering.

04
Data protection
Bucket policies, transit encryption, and KMS keys.

05
Monitoring
Cloud-native logging, audit trails, and anomaly detection.

06
Misconfiguration risks
Root causes of 80%+ cloud breaches and remediations.

07
Container security
Image hardening, runtime isolation, and Kubernetes RBAC.

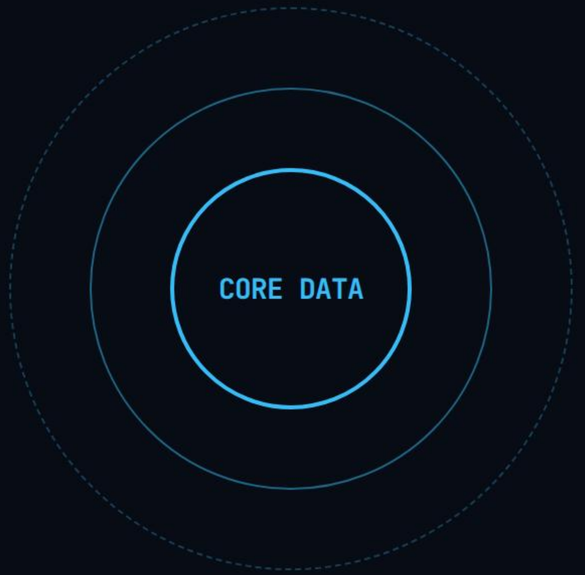
08
Compliance posture
CIS benchmarks, automated audits, and cloud governance.

Security Architecture

Designing systems where security is structural, not bolted-on — threat-modelled, layered, and resilient against evolving threats.

CONCEPTS MEMBERS WORK WITH

- Secure system design
- Zero Trust principles
- Threat modelling (STRIDE)
- Defence in depth
- Network zoning & micro-segmentation
- Identity & access frameworks
- Risk management & trade-offs
- Resilient recovery patterns



Awareness

Reading the threat landscape as it actually moves — beyond sensational headlines, into the mechanics of phishing, social engineering, malware, and emerging attack vectors.

FOCUS AREAS

- Cybersecurity news and active threat campaigns
- Online scams, spear-phishing & social engineering
- Social engineering behavioural patterns
- Malware families, ransomware, and payloads
- Digital privacy, consumer rights & emerging tech

ACTIVITIES

Weekly news briefings	RECURRING
Awareness campaigns	PUBLIC-FACING
Discussions & structured debates	MEMBER-LED
Content creation & explainers	WRITING

Explore

A platform where students try different non-technical domains, discover their interests and potential, and choose skills they want to develop further.

WHAT STUDENTS EXPLORE

- Discover interests beyond core technical tracks
- Real-world creative & communication projects
- Leadership, teamwork & event operations
- PR, networking & industry outreach opportunities

PROFESSIONAL SKILLS BUILT

Communication | Public speaking | Presentation

Leadership | Teamwork | Event management

Discover Interests

Twelve ways members show up and contribute.

Activities range from technical to communicative. Participation scales to the member — show up, run a session, mentor juniors, or lead a project.

01
Workshops
Recurring hands-on sessions.

02
Peer-to-peer learning
Members teach members.

03
Resource sharing
Curated papers and tools.

04
Technical sessions
Deep dives into tools.

05
Projects
Member-led builds spanning terms.

06
Competitions
CTFs and external contests.

07
Awareness campaigns
Outward sessions on scams.

08
Debates
Structured arguments on policy.

09
Presentations
Members report on research.

10
Career sessions
Roles, certifications, and prep.

11
Guest sessions
Practitioners and alumni speak.

12
Cyber discussions
Open conversations on field shifts.

What the club delivers — to students and to the department.

Complementary benefits on both sides — students develop as practitioners, the department builds culture and visibility.

TO STUDENTS — NINE OUTCOMES

Domain exposure	**Structured learning**	**Peer learning**
Practical experience	**Career awareness**	**Communication**
Leadership	**Competition experience**	**Project portfolios**

TO THE DEPARTMENT — SEVEN OUTCOMES

01 Strong cybersecurity community culture

02 Increased active student engagement

03 Consistent pipeline for national CTF competitions

04 Student-led workshops supporting curriculum

05 High-visibility student initiatives

06 Industry interaction and guest sessions

07 Demonstrable talent development

Five challenges we name honestly, with solutions attached.

01 / CHALLENGE: Different skill levels
Members join with widely varying technical backgrounds.

01 / RESPONSE: Faculty guidance & foundation stage
A shared starting floor before advanced tracks diverge.

02 / CHALLENGE: Maintaining participation
Engagement wanes without clear, recurring structure.

02 / RESPONSE: Defined member responsibilities
Member-owned roles with visible, predictable commitments.

03 / CHALLENGE: Quality control
Sessions and resources vary in technical accuracy.

03 / RESPONSE: Content review & peer mentorship
Senior members and faculty vet published club resources.

04 / CHALLENGE: Resource management
Tools, lab VMs, and environments take effort to maintain.

04 / RESPONSE: Structured, curated repository
A maintained living library rather than an untracked link list.

05 / CHALLENGE: Continuity
Student clubs risk fading when founders graduate.

05 / RESPONSE: Documented handover processes
Operational playbooks ensure incoming cohorts take over smoothly.